Healthcare Data Engineer Roadmap

Action Kit

Build the data pipelines, warehouses, and integration layers that power every analytics team, AI model, and reporting dashboard in healthcare. The infrastructure role that makes all other data roles possible.

Difficulty	High
Timeline	6 to 12 months
Last Updated	2026-03-25

This action kit contains your 90-day execution plan, portfolio project templates, resume bullet formulas, interview preparation questions, and resource checklist.

Built by Ehoneah Obed | thetransmutation.substack.com | quiz.ehoneahobed.com

Section 1: 90-Day Execution Plan

This plan maps directly to your learning path phases. Each month has specific deliverables and checkpoints to keep you on track.

Month 1: Foundation

Hours: 100 to 150

- Python programming fundamentals (data types, functions, file I/O, pandas, error handling)
- SQL mastery (complex joins, window functions, CTEs, stored procedures, query optimization)
- Relational database design and normalization (PostgreSQL or MySQL)
- Command line proficiency and version control with Git
- Healthcare data standards overview (HL7v2, FHIR, DICOM, ICD-10, SNOMED CT, LOINC)
- Data modeling fundamentals (star schema, snowflake, dimensional modeling)

Checkpoint: Write a Python script that reads a CSV of clinical data, validates values against expected ranges (e.g., vital signs), transforms it into a normalized schema, and loads it into a PostgreSQL database. Push the project to GitHub with proper documentation.

Month 2: Pipeline Engineering

Hours: 120 to 180

- ETL/ELT pipeline design patterns and orchestration (Apache Airflow, Prefect)
- Cloud platform fundamentals (AWS or Azure or GCP: storage, compute, networking)
- Data warehouse architecture (Snowflake, BigQuery, or Redshift)
- Data lake and medallion architecture (bronze, silver, gold layers)
- Healthcare API integration (FHIR REST APIs, HL7v2 message parsing, SMART on FHIR)
- Data quality frameworks and testing (Great Expectations, dbt tests)

Checkpoint: Build an end-to-end data pipeline using Airflow that ingests synthetic FHIR patient data from a public API, transforms it through bronze/silver/gold layers, loads it into a cloud data warehouse, and includes automated data quality checks. Deploy on a free-tier cloud account.

Month 3: Specialization

Hours: 100 to 150

- Track A: Clinical Data Infrastructure (EHR data extraction, clinical data warehouses, research data platforms, OMOP CDM)
- Track B: Payer and Claims Data Engineering (claims pipelines, member eligibility, risk adjustment, HEDIS/Stars data feeds)
- Track C: Real-Time and Streaming Data (Kafka, event-driven architecture, IoT/wearable device data, real-time clinical alerting)
- Track D: AI/ML Data Infrastructure (feature stores, model serving pipelines, training data preparation, MLOps for healthcare)

Checkpoint: Complete a cloud data engineering certification (AWS, Azure, or Databricks). Build a specialization-track portfolio project. Apply to 5 healthcare data engineer positions targeting your chosen track.

Section 2: Portfolio Project Templates

Each project below includes a dataset, tools, timeline, and your clinical advantage. Complete at least 2 to 3 of these for a competitive portfolio.

Project 1: FHIR Patient Data Pipeline

Build an end-to-end data pipeline that ingests synthetic FHIR patient data from the Synthea generator, parses FHIR Bundle resources, transforms them through bronze (raw JSON), silver (flattened, typed, validated), and gold (analytics-ready fact and dimension tables) layers, and loads them into a cloud data warehouse. Include automated data quality checks using Great Expectations.

Dataset	Synthea Synthetic Patient Data
URL	https://synthetichealth.github.io/synthea/
Tools	Python, Apache Airflow, PostgreSQL or Snowflake, Great Expectations, Docker, Git
Time Estimate	4 to 6 weeks

Clinical Advantage: You understand what FHIR patient resources represent clinically, so you can build validation rules that catch data quality issues a non-clinical engineer would miss, like a medication dosage outside therapeutic range or a lab result that is physiologically impossible.

Project 2: Claims Data Warehouse with Quality Measures

Design and build a dimensional data warehouse from CMS public claims data. Model fact and dimension tables for claims analysis. Build dbt transformation models that calculate quality measures (e.g., diabetes screening rates, preventive care utilization). Create a data dictionary and automated documentation.

Dataset	CMS Synthetic Public Use Files
URL	https://data.cms.gov/collection/synthetic-medicare-enrollment-fee-for-service-claims-and-prescription-drug
Tools	Python, dbt, PostgreSQL or BigQuery, Git, Markdown for documentation
Time Estimate	4 to 6 weeks

Clinical Advantage: You understand what quality measures actually measure clinically, so your data model captures the right grain and your transformation logic correctly handles the clinical edge cases (e.g., patients with valid exclusions, claims that span measurement years).

Project 3: Real-Time Clinical Data Stream Processor

Build a streaming data pipeline that simulates real-time vital signs data from patient monitoring devices, processes the stream to detect anomalies (critical values, rapid deterioration patterns), and writes alerts to a dashboard. Use Kafka or a cloud streaming service for the messaging layer.

Dataset	MIMIC-III Clinical Database (PhysioNet)
URL	https://physionet.org/content/mimiciii/1.4/
Tools	Python, Apache Kafka or AWS Kinesis, InfluxDB or TimescaleDB, Grafana for dashboarding

Time Estimate	3 to 5 weeks

Clinical Advantage: You know what constitutes a clinically significant vital sign change versus normal variation. Your anomaly detection thresholds will be clinically meaningful rather than purely statistical, reducing false alerts.

Project 4: Healthcare Data Governance and Lineage Platform

Build a data catalog and lineage tracking system for a multi-source healthcare data environment. Implement automated PHI detection and classification, document data lineage from source to consumption, and create a metadata management layer that tracks data freshness, ownership, and quality scores.

Dataset	Synthea combined with CMS public data
URL	https://synthetichealth.github.io/synthea/
Tools	Python, OpenMetadata or Apache Atlas, PostgreSQL, dbt for lineage, Git
Time Estimate	3 to 4 weeks

Clinical Advantage: Your HIPAA training and clinical experience mean you can build PHI detection rules that go beyond simple pattern matching. You understand that a combination of date, zip code, and diagnosis can re-identify a patient even when no single field is obviously PHI.

Project 5: EHR-to-OMOP Data Transformation Pipeline

Build a pipeline that transforms raw EHR-format data into the OMOP Common Data Model used by 300+ research institutions globally. Map source codes (ICD-10, RxNorm, LOINC) to OMOP standard concepts. Include vocabulary management and concept mapping validation.

Dataset	Synthea FHIR data mapped to OMOP
URL	https://ohdsi.github.io/CommonDataModel/
Tools	Python, SQL, OHDSI tools (Usagi, Achilles), PostgreSQL, dbt
Time Estimate	4 to 6 weeks

Clinical Advantage: You understand the clinical meaning behind ICD-10 codes, medication classifications, and lab result interpretations. This means your concept mappings will be clinically accurate, not just string matches, catching the subtle distinctions that determine research validity.

Section 3: Resume Bullet Formulas

Use these formulas to translate your clinical experience into language that resonates with hiring managers in health data roles. Replace bracketed text with your specifics.

Nursing Background

• Leveraged [clinical skill: EHR documentation and clinical data entry across multiple modules] to deliver [tech outcome: Source system knowledge and data model understanding], resulting in [quantified impact].

• Leveraged [clinical skill: Shift report handoffs and information continuity protocols] to deliver [tech outcome: Data pipeline reliability and data lineage tracking], resulting in [quantified impact].

• Leveraged [clinical skill: Quality metric tracking and regulatory reporting (Core Measures, HCAHPS)] to deliver [tech outcome: ETL logic design and reporting pipeline construction], resulting in [quantified impact].

• Leveraged [clinical skill: Medication reconciliation across care transitions] to deliver [tech outcome: Data reconciliation and deduplication logic], resulting in [quantified impact].

Pharmacy Background

• Leveraged [clinical skill: Drug interaction checking and clinical decision support systems] to deliver [tech outcome: Data validation rules and business logic implementation], resulting in [quantified impact].

• Leveraged [clinical skill: Formulary management and medication database maintenance] to deliver [tech outcome: Reference data management and master data governance], resulting in [quantified impact].

• Leveraged [clinical skill: 340B program compliance tracking and reporting] to deliver [tech outcome: Regulatory compliance pipeline design and audit trail construction], resulting in [quantified impact].

• Leveraged [clinical skill: Automated dispensing cabinet management and inventory optimization] to deliver [tech outcome: System integration and real-time data flow management], resulting in [quantified impact].

Other Clinical Background

• Leveraged [clinical skill: Laboratory information system (LIS) management and result reporting] to deliver [tech outcome: Data integration and HL7/FHIR message processing], resulting in [quantified impact].

• Leveraged [clinical skill: Diagnostic imaging workflow and PACS system management] to deliver [tech outcome: Large-scale data storage, retrieval, and metadata management], resulting in [quantified impact].

• Leveraged [clinical skill: Clinical research data collection and CRF management] to deliver [tech outcome: Data schema design and structured data collection pipelines], resulting in [quantified impact].

• Leveraged [clinical skill: Revenue cycle and coding workflow management] to deliver [tech outcome: Transaction processing pipeline design], resulting in [quantified impact].

Pro Tip: Always quantify. Instead of 'analyzed data,' write 'analyzed 50,000+ claims records to identify $2.3M in recoverable overpayments.'

Section 4: Interview Preparation Questions

Practice these role-specific questions. For each, prepare a 2-minute answer that highlights both your technical skills and clinical background.

Technical Questions

1. Walk me through how you would design a data pipeline to ingest HL7v2 ADT messages from a hospital's EHR and load them into a cloud data warehouse in near real-time.
2. You are building a medallion architecture (bronze, silver, gold) for healthcare claims data. Describe what transformations happen at each layer and why.
3. How would you parse and flatten FHIR Bundle resources containing Patient, Condition, and MedicationRequest entries into relational tables suitable for analytics?
4. Describe your approach to implementing data quality checks in a healthcare ETL pipeline. What clinical data validation rules would you build that a non-clinical engineer would miss?
5. A data pipeline processing lab results is running 4 hours behind schedule. Walk me through how you would diagnose the bottleneck and optimize performance.

Behavioral Questions

1. Tell me about a time your clinical background helped you catch a data quality issue or build a better validation rule than a purely technical engineer would have.
2. Describe how you would explain a complex data architecture decision (like choosing between batch and streaming processing) to a clinical stakeholder who needs to understand the trade-offs.
3. How do you balance the pressure to deliver pipelines quickly against the need for thorough data quality and PHI compliance in healthcare?
4. Tell me about a time you had to learn a new technical tool or framework on a tight timeline. How did you approach it?

Scenario Questions

1. A downstream analytics team reports that patient counts in your data warehouse do not match the source EHR system. The discrepancy is about 3%. How do you investigate and resolve this?
2. You discover that a data feed from a lab vendor has been sending duplicate results for the past two weeks. Some of these duplicates have already been consumed by a machine learning model in production. What do you do?
3. Your organization is migrating from an on-premises data warehouse to a cloud-based lakehouse. The CISO is concerned about PHI in the cloud. How do you design the pipeline to address these concerns?

Section 5: Resource Checklist

Use this checklist to track your progress through all recommended resources.

Certifications

Certification	Signal	Cost	Timeline
AWS Certified Data Engineer, Associate	High	$150 exam fee	2 to 4 months study
Databricks Certified Data Engineer Associate	High	$200 exam fee	2 to 3 months study
Azure Data Engineer Associate (DP-700)	High	$165 exam fee	2 to 4 months study
Google Cloud Professional Data Engineer	Helpful	$200 exam fee	3 to 4 months study
CHDA (Certified Health Data Analyst, AHIMA)	Helpful	$259 to $329 exam fee	2 to 3 months study, requires healthcare data experience
dbt Analytics Engineering Certification	Helpful	Free	1 to 2 months practice
Snowflake SnowPro Core Certification	Helpful	$175 exam fee	1 to 2 months study

Learning Resources by Phase

Foundation

- [Paid] DataCamp Data Engineer Track: https://www.datacamp.com/tracks/data-engineer
- [Free] Mode SQL Tutorial: https://mode.com/sql-tutorial/
- [Free] Python for Everybody (University of Michigan): https://www.py4e.com/
- [Free] HL7 FHIR Official Documentation: https://hl7.org/fhir/
- [Paid] Designing Data-Intensive Applications (Martin Kleppmann): https://www.oreilly.com/library/view/designing-data-intensive-applications/9781491903063/
- [Free] GitHub Learning Lab: https://skills.github.com/

Pipeline Engineering

- [Free] Apache Airflow Documentation and Tutorials: https://airflow.apache.org/docs/apache-airflow/stable/tutorial/index.html
- [Free] AWS Cloud Practitioner Essentials: https://aws.amazon.com/training/learn-about/cloud-practitioner/
- [Free] dbt (data build tool) Documentation: https://docs.getdbt.com/docs/introduction
- [Free] Snowflake University: https://learn.snowflake.com/en/
- [Free] SMART on FHIR Documentation: https://docs.smarthealthit.org/

• [Free] Great Expectations Documentation: https://docs.greatexpectations.io/docs/

Specialization

• [Free] OHDSI OMOP Common Data Model: https://ohdsi.github.io/CommonDataModel/

• [Free] Synthea Synthetic Patient Generator: https://synthetichealth.github.io/synthea/

• [Paid] AWS Certified Data Engineer Associate Prep: https://aws.amazon.com/certification/certified-data-engineer-associate/

• [Free] Databricks Academy: https://www.databricks.com/learn/training

• [Free] Confluent Kafka Tutorials: https://developer.confluent.io/tutorials/

Your First Three Moves

Set up your development environment and start learning Python and SQL (3 hours)

• Install Python 3, VS Code (with Python extension), PostgreSQL, and Git on your computer. Create a GitHub account if you do not have one.

• Complete the first 3 modules of Python for Everybody (py4e.com) or the first section of DataCamp's Python for Data Engineering track.

• Write your first SQL queries against a sample healthcare dataset: download CMS public data, load it into PostgreSQL, and run 5 queries that answer questions you would have asked in your clinical role.

Map your clinical data knowledge and explore the healthcare data engineering landscape (2 hours)

• List every data system you touch in your clinical work: EHR modules, lab systems, pharmacy systems, reporting dashboards, quality databases. For each, note what data goes in and what comes out.

• Search LinkedIn for 'Healthcare Data Engineer' and note the top 15 employers, required skills, and common tools. Identify which cloud platform (AWS, Azure, GCP) appears most frequently.

• Read 3 articles on healthcare data interoperability (start with HL7 FHIR documentation overview at hl7.org/fhir) to understand the standards that healthcare data pipelines must support.

Build a daily coding practice and join the data engineering community (Ongoing (1 to 2 hours per day))

• Commit to 1 hour of Python or SQL practice daily. Use LeetCode (SQL problems), HackerRank (Python challenges), or Mode SQL Tutorial for structured practice.

• Join the Data Engineering subreddit (r/dataengineering), dbt Community Slack, and Healthcare Data Analytics LinkedIn groups to learn from practitioners.

• Start a GitHub repository called 'healthcare-data-engineering-portfolio' and push something to it at least once per week, even if it is a small script or SQL query.

Ready to go deeper? Take the free career quiz at quiz.ehoneahobed.com or subscribe to The Transmutation newsletter at thetransmutation.substack.com